

```
import warnings
warnings.filterwarnings('ignore')
import pandas as pd
import numpy as np
```

Data Preparation

```
# Loading the dataset
ratings= pd.read_csv(r'C:\Users\oluwa\Desktop\HNG INTERNSHIP\STAGE 1\
ml-latest-small\ratings.csv')
movies = pd.read_csv(r'C:\Users\oluwa\Desktop\HNG INTERNSHIP\STAGE 1\
ml-latest-small\movies.csv')
tags = pd.read_csv(r'C:\Users\oluwa\Desktop\HNG INTERNSHIP\STAGE 1\ml-
latest-small\tags.csv')
links = pd.read_csv(r'C:\Users\oluwa\Desktop\HNG INTERNSHIP\STAGE 1\
ml-latest-small\links.csv')

# Merging rating and movie dataset using the inner merging type
df = ratings.merge(movies, how='inner', on= 'movieId')

# Checking for duplicates and missing values
df.duplicated().sum()

# Checking for missing values
df.isnull().sum()

# Converting timestamp to datetime format
df['timestamp'] = pd.to_datetime(df['timestamp'], unit='s')
```

Feature Engineering

```
# Extracting and creating a new column containing the release year of
the movies
year = r'(\d{4})'

df['release_year']= df['title'].str.extract(year)
df['release_year']= pd.to_numeric(df['release_year']).astype('Int64')

# Counting the number of genres for each movie and creating a new
column for it
df['genre_count'] = df['genres'].str.count('\|') + 1
df.loc[df['genres'] == '(no genres listed)', 'genre_count'] = 0

print("Genre Count Value Distribution (Top 5):")
print(df['genre_count'].value_counts().sort_index().head())
```

```

# Creating a new column containing the year the rating was given
df['rating_year'] = df['timestamp'].dt.year

# Creating a new column for the users average rating
df['user_avg_rating'] = df.groupby('userId')
['rating'].transform('mean')

# Creating a new column containing the total number of rating a movie
has received
df['movie_rating_count'] = df.groupby('movieId')
['rating'].transform('count')

# Creating a new column calculating the amount of years since the movie
was released till date of rating
df['movie_age_at_rating'] = df['rating_year'] - df['release_year']

```

Exploratory Data Analysis

```

import matplotlib.pyplot as plt
import seaborn as sns

# Understanding the overall rating distribution across all movies (Do
users tend to rate movies high, mid, or low?)

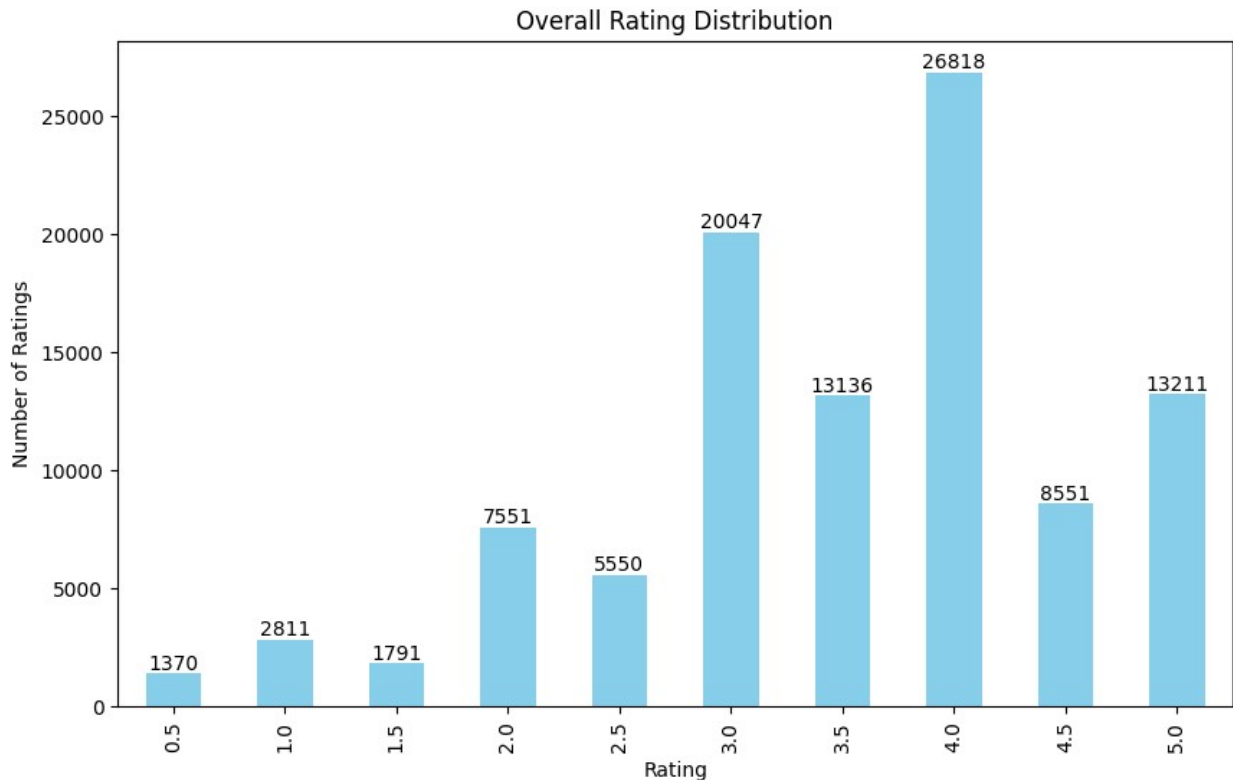
overall_rating = df['rating'].value_counts().sort_index()
overall_mean_rating = df['rating'].mean()

print(f"Overall Mean Rating is:\n{overall_mean_rating}")

# Visualizing the overall rating distribution
plt.figure(figsize=(10,6))
overall_rating.plot(kind='bar', color='skyblue')
plt.bar_label(plt.gca().containers[0])
plt.title('Overall Rating Distribution')
plt.xlabel('Rating')
plt.ylabel('Number of Ratings')
plt.show()

Overall Mean Rating is:
3.501556983616962

```



Analyzing the impact of movie popularity on ratings (Do more popular movies(those with a higher rating count) receive higher average ratings?)

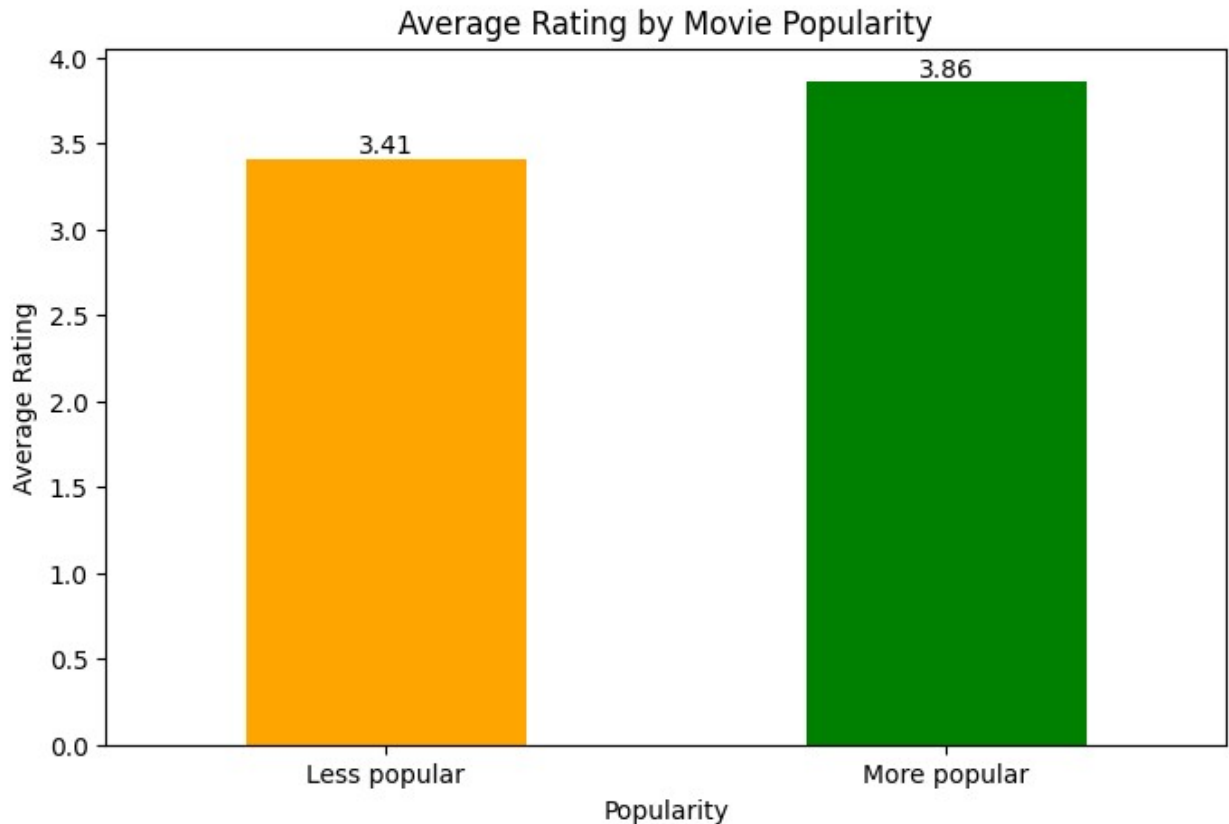
```
df['popularity'] = np.where(df['movie_rating_count'] < 100, 'Less popular', 'More popular')
```

Calculating average rating for low and high popularity movies

```
popularity_rating = df.groupby('popularity')['rating'].mean()  
popularity_rating
```

Visualizing the average rating by movie popularity

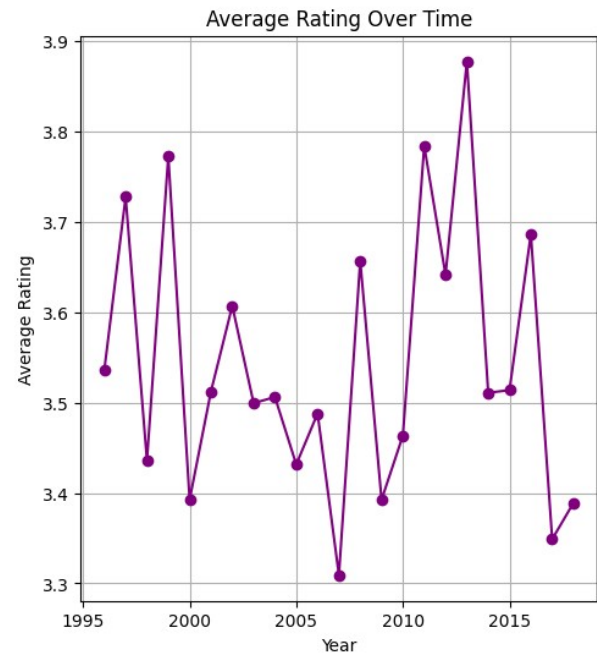
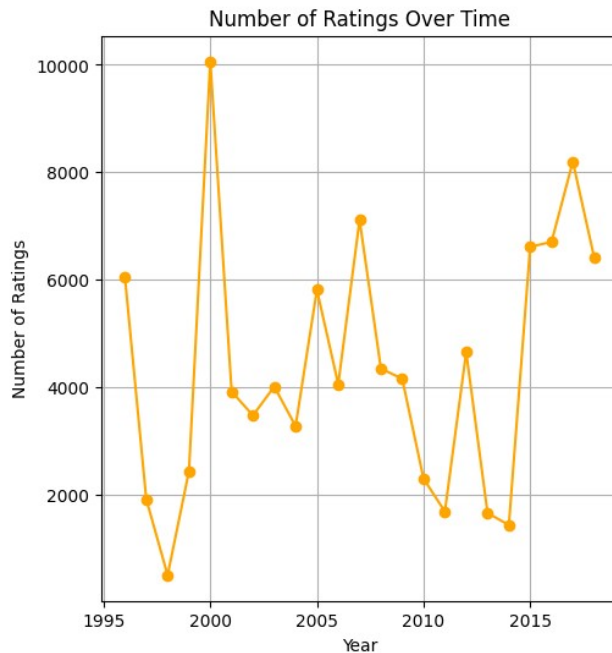
```
plt.figure(figsize=(8,5))  
popularity_rating.plot(kind='bar', color=['orange', 'green'])  
plt.bar_label(plt.gca().containers[0], fmt='%.2f')  
plt.title('Average Rating by Movie Popularity')  
plt.xlabel('Popularity')  
plt.ylabel('Average Rating')  
plt.xticks(rotation=0)  
plt.show()
```



```
# Understanding the volume of rating and average rating over time (Do users tend to give higher or lower ratings as time progresses?)  
df.groupby(df['rating_year']).agg({'rating': ['count', 'mean']})
```

```
# Plotting the number of ratings and average rating over time  
plt.figure(figsize=(12,6))  
plt.subplot(1, 2, 1)  
ratings_count = df.groupby('rating_year')['rating'].count()  
ratings_count.plot(marker='o', color='orange')  
plt.title('Number of Ratings Over Time')  
plt.xlabel('Year')  
plt.ylabel('Number of Ratings')  
plt.grid()
```

```
plt.subplot(1, 2, 2)  
ratings_over_time = df.groupby('rating_year')['rating'].mean()  
ratings_over_time.plot(marker='o', color='purple')  
plt.title('Average Rating Over Time')  
plt.xlabel('Year')  
plt.ylabel('Average Rating')  
plt.grid()  
  
plt.show()
```



```
# Insight on the genre that gets the most ratings and the genre that
# gets the highest average ratings (Popularity and Quality perception)
df_genres = df['genres'].str.split('|',
expand=True).stack().reset_index(level=1, drop=True).to_frame('genre')
df_new = df.merge(df_genres, left_index=True, right_index=True)
genre_popularity = df_new.groupby('genre').agg(rating_count
=('rating', 'count'), average_rating = ('rating', 'mean')).reset_index()

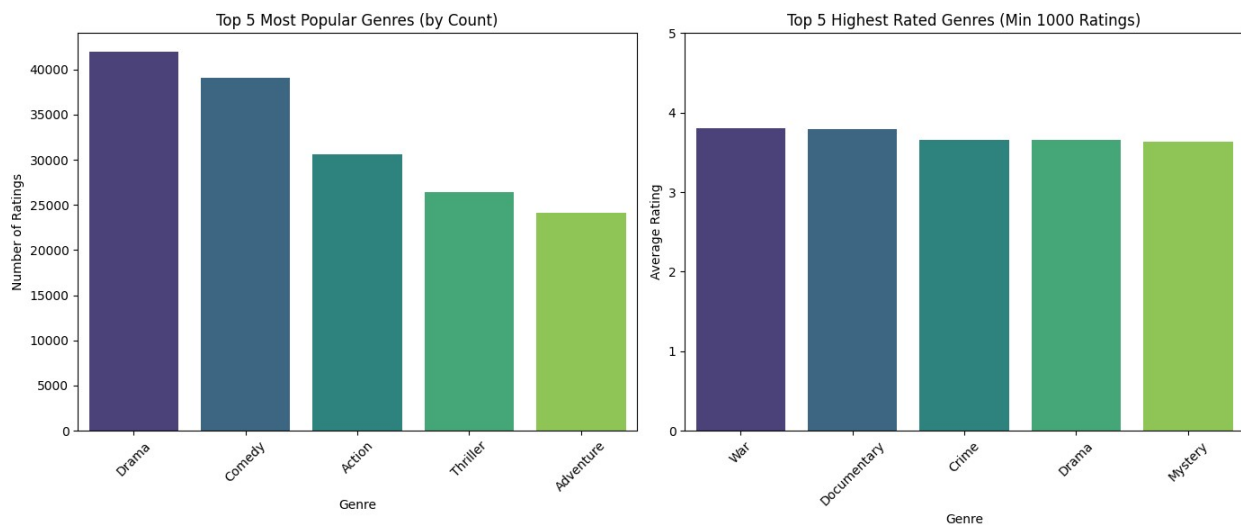
# Finding the genre with the most ratings (Popularity)
top_genres_popularity =
genre_popularity.sort_values(by='rating_count',
ascending=False).head(5)

# Finding the genre with the most average rating (Quality perception)
min_ratings = 1000
top_genres_quality = genre_popularity[genre_popularity['rating_count']
>= min_ratings].sort_values(by='average_rating',
ascending=False).head(5)

# Visualizing the results
plt.figure(figsize=(14,6))
plt.subplot(1, 2, 1)
sns.barplot(data=top_genres_popularity, x='genre', y='rating_count',
palette='viridis')
plt.title('Top 5 Most Popular Genres (by Count)')
plt.xlabel('Genre')
plt.ylabel('Number of Ratings')
plt.xticks(rotation=45)
```

```
# Visualizing the results
plt.subplot(1, 2, 2)
sns.barplot(data=top_genres_quality, x='genre', y='average_rating',
palette='viridis')
plt.title('Top 5 Highest Rated Genres (Min 1000 Ratings)')
plt.xlabel('Genre')
plt.ylabel('Average Rating')
plt.xticks(rotation=45)

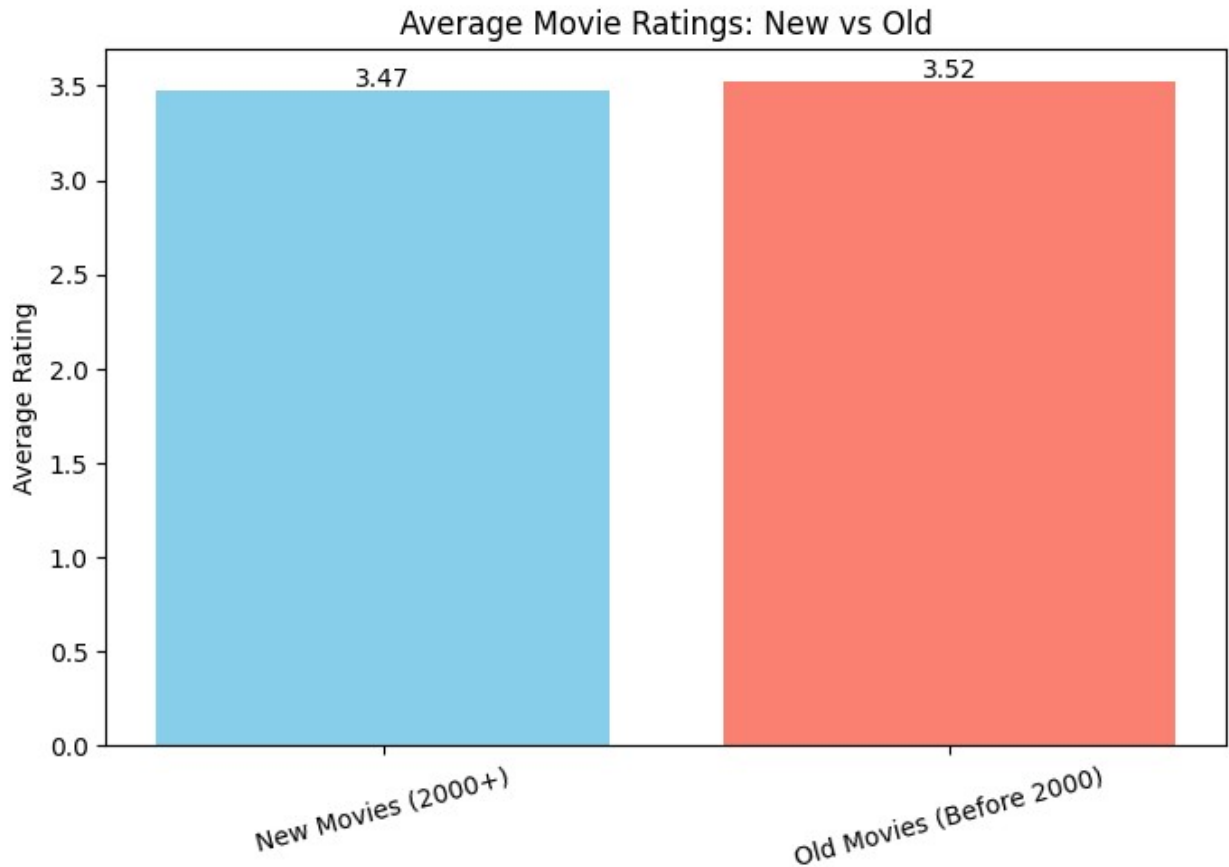
plt.ylim(0, 5)
plt.tight_layout()
plt.show()
```



```
# Analysing the impact of movie age on ratings (Do older movies tend
to receive higher or lower ratings compared to newer releases?)
new_movie = df[df['release_year'] >= 2000]['rating'].mean()

old_movie = df[df['release_year'] < 2000]['rating'].mean()

plt.figure(figsize=(8,5))
bars = plt.bar(['New Movies (2000+)', 'Old Movies (Before 2000)'],
[new_movie, old_movie], color=['skyblue', 'salmon'])
plt.bar_label(bars, fmt='%.2f')
plt.title('Average Movie Ratings: New vs Old')
plt.ylabel('Average Rating')
plt.xticks(rotation=15)
plt.show()
```

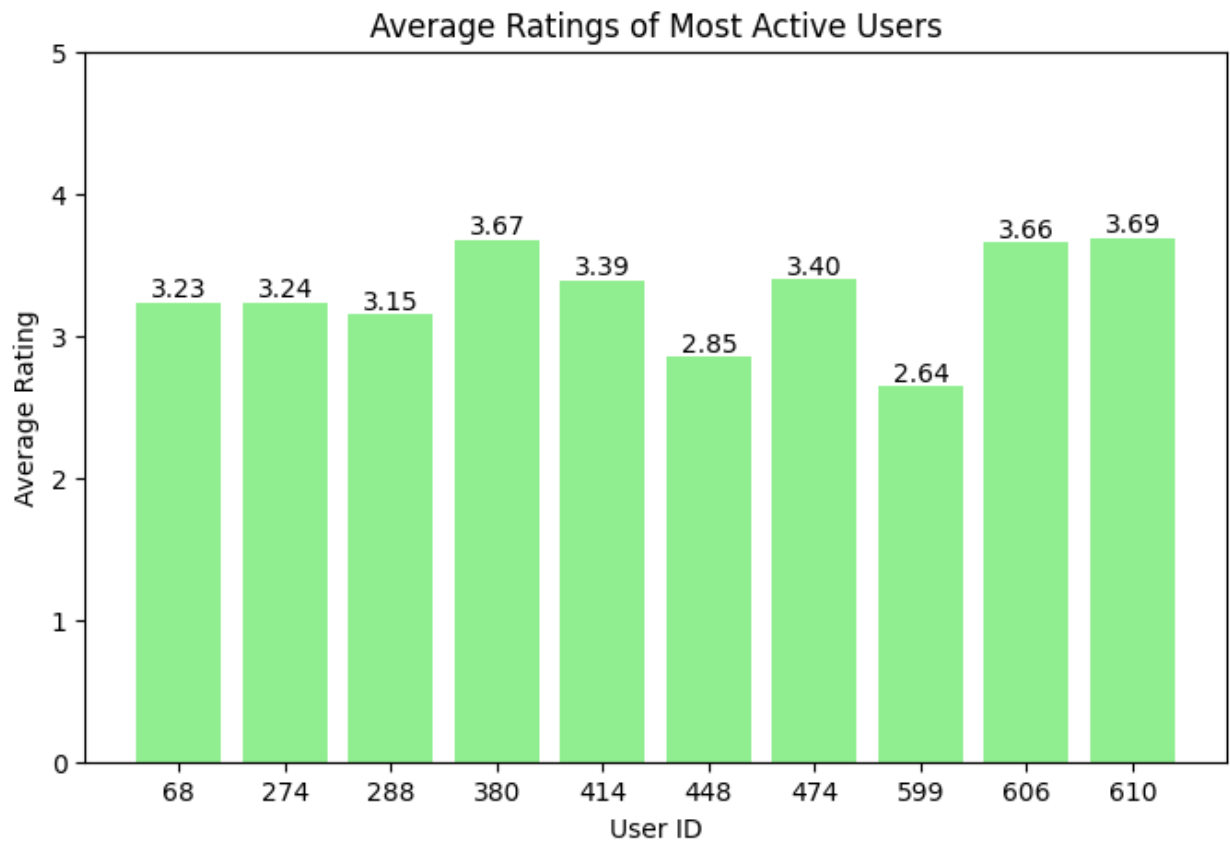


Identifying the most active users and their rating behavior (Do the most active users tend to rate movies higher or lower than the average user?)

```
user_activity = df['userId'].value_counts().head(10).index
active_users = df[df['userId'].isin(user_activity)]
active_user_ratings = active_users.groupby('userId')['rating'].mean()
active_user_ratings
```

Visualizing the average ratings of the most active users

```
plt.figure(figsize=(8,5))
bars = plt.bar(active_user_ratings.index.astype(str),
               active_user_ratings.values, color='lightgreen')
plt.title('Average Ratings of Most Active Users')
plt.xlabel('User ID')
plt.ylabel('Average Rating')
plt.ylim(0, 5)
plt.bar_label(bars, fmt='%.2f')
plt.show()
```



```
df.to_csv("cleaned_stage1.csv", index=False)
```